

WORLDFORGE

System Architecture & Knowledge Model

Version 0.1 — Arizona Explorer Foundation

Learn by Living the Story.

Every Place Has a Story.

Document owner	WorldForge
Status	Draft for architecture review
Primary scope	Arizona Explorer MVP
Architecture principle	Accessibility is architecture
Knowledge principle	Everything is connected

1. Executive Summary

This document defines the technical and conceptual foundation for WorldForge, beginning with the Arizona Explorer MVP. It translates the product vision into a buildable system: a web platform that connects places, people, species, organizations, artifacts, events, timelines, stories, learning activities, and career pathways through a shared knowledge model.

The architecture is intentionally modular. The first release can run on a focused web stack using Next.js, FastAPI, PostgreSQL, object storage, map services, and retrieval-augmented AI. More complex graph, 3D, mobile, multiplayer, and immersive capabilities are introduced only when the product demonstrates demand and the data model requires them.

WorldForge is not designed as a collection of disconnected pages. It is designed as a connected civilization system. A user should be able to move from the Grand Canyon to the Colorado River, from the river to geology, from geology to Indigenous history, from history to artifacts and museums, and from those subjects to lessons, skills, careers, and creative projects.

2. Architectural Goals and Principles

Principle	Architectural consequence
Connected by design	All major entities participate in explicit, typed relationships. Pages are views over a connected model, not isolated content records.
Evidence before confidence	Factual claims should be supported by sources, provenance, review status, and confidence labels.
Fact, tradition, interpretation, and fiction remain distinguishable	The system must label historical evidence, living traditions, mythology, interpretation, adaptation, and fictional invention separately.
Accessibility is architecture	Keyboard access, semantic structure, captions, transcripts, screen-reader support, reduced motion, language readiness, and multiple presentation modes are platform requirements.
Start simple, scale deliberately	PostgreSQL supports the MVP. Dedicated graph and search systems are added when demonstrated needs justify operational complexity.
Content is reusable	A place, event, species, or artifact can appear in maps, timelines, quests, AI responses, exhibits, and future immersive experiences without duplication.
Licensing is explicit	Every content item has ownership, usage rights, license, and permitted experience types.
Human review remains possible	Curators, educators, community partners, and subject experts can review, correct, approve, and annotate content.

3. System Context

WorldForge serves explorers, learners, educators, families, creators, curators, cultural partners, and future employers. The MVP focuses on explorers and educators while preserving extension points for institutional publishing and creator tools.

3.1 External Actors

- Explorer: browses maps, pages, timelines, and connected topics; asks the AI Guide questions; completes quests.
- Educator: assembles journeys, assigns quests, reviews learner progress, and recommends resources.
- Curator or subject expert: creates and reviews content, sources, relationships, and interpretive notes.
- Cultural or community partner: controls representation guidance, preferred terminology, access restrictions, and corrections for community-associated content.
- Administrator: manages users, permissions, moderation, licensing records, and system health.
- External services: map tiles/geocoding, AI models, speech, object storage, analytics, and identity providers.

3.2 System Boundary

Inside WorldForge	Integrated service	Not owned by WorldForge
Entity and relationship data	Map rendering/geocoding	Third-party base-map data
Editorial content and citations	AI model inference	External model training data
Quests, passports, and progress	Email/identity provider	User email infrastructure
AI retrieval and response policies	Speech and transcription	Device operating systems
Rights and provenance records	Media delivery/CDN	Partner-owned copyrighted works

4. Logical Architecture

4.1 Layered Model

Layer	Primary responsibilities	MVP implementation
Experience	Web interface, accessible navigation, map, timeline, pages, passport	Next.js, React, TypeScript
Application	Authentication, journeys, quests, favorites, progress, permissions	FastAPI services
Knowledge	Entities, relationships, classifications, sources, spatial and temporal data	PostgreSQL with relational graph tables
AI orchestration	Question interpretation, retrieval, grounding, citations, response formatting	Python service with OpenAI API and policy layer
Search and discovery	Keyword, semantic, geographic, temporal, and relationship traversal	PostgreSQL full-text + vector extension initially
Media	Images, audio, transcripts,	Object storage and CDN

	documents, thumbnails	
Operations	Logging, analytics, moderation, quality review, backups	Cloud monitoring and administration tools

4.2 Recommended MVP Components

- WorldForge Web: server-rendered, responsive, accessible user experience.
- WorldForge API: REST/JSON endpoints for content, map, timeline, quests, progress, and administration.
- Knowledge Service: entity retrieval, typed relationship traversal, filtering, and citation lookup.
- AI Guide Service: grounded question answering, explanation modes, safety checks, and citation assembly.
- Ingestion Pipeline: validates metadata, imports partner or editorial content, generates embeddings, and records provenance.
- Editorial Console: basic content review, source management, relationship editing, and publication workflow.
- Analytics and Audit Service: privacy-conscious product events, AI traces, change history, and quality reports.

5. Core Knowledge Model

The core model uses entities, relationships, classifications, sources, and presentations. Entities represent things that exist or are discussed. Relationships explain how they connect. Classifications provide controlled vocabulary. Sources support claims. Presentations adapt the same knowledge for different users and experiences.

5.1 Root Entity Types

Entity type	Definition
World	A coherent reality or setting: Earth, a historical simulation, an original world, or a licensed fictional universe.
Place	A spatially located or conceptually bounded location: region, city, park, building, room, biome, planet, or starship.
Person or Character	A real person, historical person, fictional character, collective persona, or guided interpretation role.
Organization	A government, nation, school, museum, business, emergency service, community group, guild, or fictional institution.
Event	A bounded occurrence with time, participants, causes, consequences, places, and evidence.
Species	A biological, mythical, fictional, or alien species, with explicit classification type.
Artifact	A physical or digital object, artwork, tool, vehicle, garment, manuscript, monument, or fictional object.

Story	A book, oral tradition, film, episode, comic, game, legend, adaptation, or user-created narrative.
Concept	A scientific, cultural, linguistic, technical, artistic, legal, or educational idea.
Timeline	An ordered lens across events and periods; may be historical, natural, personal, organizational, or fictional.
Learning Activity	A quest, lesson, reflection, experiment, field guide, project, or assessment.
Skill or Career	A competency, role, occupation, training pathway, credential, internship, or employment opportunity.
Source	A publication, dataset, interview, archival record, partner statement, or media item used as evidence.

5.2 Universal Entity Fields

- **id:** stable internal identifier, preferably UUID.
- **slug:** human-readable URL identifier, unique within a scope.
- **entity_type** and **subtype:** root type plus controlled subtype.
- **name**, **short_name**, **alternate_names**, and **pronunciation**.
- **summary** and **structured description blocks**.
- **world_id** and **canonical scope**.
- **status:** draft, review, published, archived, disputed, or restricted.
- **fact_class:** documented fact, scholarly interpretation, community tradition, mythology, fiction, adaptation, simulation, or user-created.
- **rights_profile_id:** owner, license, permitted uses, attribution, and expiration.
- **access_profile_id:** public, age-gated, classroom-only, partner-only, culturally restricted, or private.
- **created_at**, **updated_at**, **published_at**, **created_by**, and **revision number**.
- **accessibility metadata:** plain-language summary, pronunciation, transcript, audio description, sign-language asset, sensory notes, and reading level.

6. Entity Specifications

6.1 World

Key structured fields: **world_type**, **canonical_status**, **time_system**, **geographic_system**, **rights_scope**, **default_fact_class**, **entry_place_id**.

6.2 Place

Key structured fields: **place_type**, **parent_place_id**, **geometry**, **latitude/longitude**, **elevation**, **address**, **timezone**, **accessibility_features**, **visitor_conditions**.

6.3 Person or Character

Key structured fields: **person_kind**, **birth/debut data**, **heritage descriptors**, **languages**, **occupations**, **abilities**, **accessibility and health representation**, **privacy level**.

6.4 Organization

Key structured fields: organization_type, jurisdiction, founded/dissolved dates, headquarters, memberships, services, governance model.

6.5 Event

Key structured fields: event_type, start/end or uncertain range, locations, participants, causes, outcomes, evidence state, recurrence.

6.6 Species

Key structured fields: classification_system, native range, habitats, conservation status, domestication/working roles, mythical or fictional taxonomy.

6.7 Artifact

Key structured fields: artifact_type, materials, creator/manufacturer, creation period, current location, ownership history, dimensions, condition, rights.

6.8 Story

Key structured fields: story_type, creator, publication/release data, canon scope, setting, characters, adaptations, content guidance, rights.

6.9 Learning Activity

Key structured fields: audience, learning objectives, estimated duration, prerequisites, accessibility variants, assessment method, skills earned.

6.10 Skill or Career

Key structured fields: skill framework, proficiency level, evidence requirements, related activities, education pathway, employers/industries, opportunities.

7. Relationship Model

Relationships are first-class records. Each relationship has a type, direction, validity period, source support, confidence, fact class, and optional explanatory note. This permits WorldForge to represent change over time and disagreement without overwriting history.

7.1 Relationship Record

- subject_entity_id, predicate_type, object_entity_id.
- directionality: directed, symmetric, or inverse pair.
- valid_from and valid_to, including uncertain or approximate dates.
- world/canon scope.
- fact class and confidence.
- source links and supporting claim notes.
- editorial status and reviewer.
- visibility and cultural access rules.

7.2 High-Value Relationship Types

Relationship	Examples
contains / located_in	World → Place; Place → Place; Artifact → Place
participated_in	Person/Organization → Event
occurred_at	Event → Place
member_of / operates	Person → Organization; Organization → Place
created / authored / manufactured	Person or Organization → Artifact or Story
depicts / mentions / set_in	Story or Artifact → Entity
inhabits / native_to / observed_at	Species → Place
uses / made_of / derived_from	Artifact or Product → Concept, Species, or Material
caused / influenced / followed_by	Event → Event
adaptation_of / inspired_by	Story → Story, Mythology, Event, or Person
teaches / practices / evidences	Learning Activity → Concept or Skill
prepares_for / required_by	Skill → Career; Activity → Skill
contrasts_with / disputed_by	Claim or interpretation relationship

8. Timeline and Event Architecture

WorldForge timelines are generated views, not duplicated narratives. The same event can appear in a natural-history timeline, a state-history timeline, a community timeline, a character timeline, or a story canon timeline depending on filters and interpretation.

- Precise time: timestamp or date.
- Range: start and end.
- Approximate time: circa date or century.
- Uncertain interval: earliest and latest possible dates.
- Relative sequence: before, after, overlaps, during.
- Recurring event: seasonal, annual, or rule-based.
- Alternative chronology: competing scholarly or fictional canon sequences.

8.1 Timeline Views

- Natural history
- Indigenous and community histories
- Political and civic history
- Technology and infrastructure
- Species and conservation
- Organization or institution
- Person or character
- Story canon and adaptations
- Learning journey

9. Content, Evidence, and Trust

9.1 Claim-Level Evidence

A page may contain many claims with different evidence. Important factual assertions should therefore be represented as structured claims linked to sources rather than relying only on page-level bibliographies.

- Claim text or structured assertion.
- Subject, predicate, and value/entity.
- Source and page/time location.
- Quotation only when permitted and necessary.
- Confidence and review state.
- Fact class and interpretation label.
- Community note or dispute note.
- Last verification date.

9.2 Source Quality Levels

Level	Meaning
A	Primary evidence, official records, peer-reviewed research, direct partner authority
B	Reputable scholarly or institutional synthesis
C	Quality journalism, professional reference, or established secondary source
D	Community contribution awaiting review
E	Unverified, fictional, speculative, or intentionally user-created content

9.3 Publication Workflow

1. Create or import draft content.
2. Validate required metadata and rights profile.
3. Attach sources and classify claims.
4. Run automated checks for broken links, missing citations, inaccessible media, and prohibited content.
5. Route to appropriate reviewer: editorial, subject expert, accessibility, legal, or community partner.
6. Publish with revision history.
7. Monitor corrections, disputes, source changes, and license expiration.

10. AI Guide Architecture

The AI Guide is an orchestrated retrieval and presentation system. It should answer from WorldForge content and approved external sources, show citations, label uncertainty, and avoid presenting fictional or interpretive material as established fact.

10.1 Response Pipeline

8. Detect user intent, audience level, location context, and accessibility preferences.
9. Classify the question as factual, interpretive, creative, navigational, or safety-sensitive.

10. Retrieve relevant entities, claims, relationships, sources, and permitted media.
11. Rank evidence by relevance, authority, freshness, and rights constraints.
12. Generate a grounded answer using a defined explanation role such as ranger, historian, curator, or career guide.
13. Attach citations and fact-class labels.
14. Run safety, cultural, licensing, and hallucination checks.
15. Record an auditable trace without retaining unnecessary personal content.

10.2 Guide Modes

- Quick answer
- Guided tour
- Ask a ranger/curator/teacher
- Kid-friendly explanation
- Plain-language explanation
- Audio-first exploration
- Career connection
- Compare fact and fiction
- Creative companion using authorized content

10.3 AI Refusal and Escalation

The guide should decline to invent citations, disclose restricted cultural knowledge, provide unlicensed reproductions, or state uncertain claims as settled fact. It should offer safe summaries, direct users to approved sources, or flag a question for human review when necessary.

11. Search and Discovery

Discovery combines text relevance, geography, time, relationships, audience, and learning goals. A user may search directly, browse a map, move along a timeline, follow a relationship, or accept an AI-guided recommendation.

- Keyword and phrase search across names, summaries, aliases, and source metadata.
- Semantic retrieval for conceptually related material.
- Geospatial search by map bounds, radius, route, region, or accessibility.
- Temporal filtering by date, period, sequence, or uncertainty range.
- Graph traversal by relationship type and depth.
- Facets for fact class, content type, age level, language, accessibility format, and rights availability.
- Editorially designed journeys that combine entities into meaningful routes.

12. Accessibility Architecture

WorldForge must support multiple ways of perceiving, navigating, understanding, and contributing. Accessibility preferences should be stored as optional user-controlled settings and applied consistently across pages, maps, timelines, media, quests, and AI responses.

Area	Requirement
Structure	Semantic HTML, logical headings, landmarks, labels, focus order, and skip navigation.
Input	Keyboard, touch, switch-compatible controls, voice-ready actions, and generous target sizes.
Visual	Text resizing, reflow, contrast, non-color cues, reduced motion, zoom-safe layouts.
Audio	Captions, transcripts, volume controls, no audio-only instructions.
Nonvisual	Alt text, long descriptions, map alternatives, data tables, spatial summaries.
Cognitive	Plain-language mode, consistent navigation, step-by-step journeys, pause/resume, reduced distractions.
Language	Translation-ready content, pronunciation, language tags, right-to-left readiness.
Representation	Respectful disability representation and avoidance of compulsory movement metaphors in required instructions.

13. Security, Privacy, and Governance

- Collect the minimum personal data needed for accounts, saved progress, and educator features.
- Separate child, learner, educator, curator, partner, and administrator permissions.
- Encrypt data in transit and at rest; rotate secrets; audit privileged actions.
- Use privacy-conscious analytics and clear retention periods.
- Provide deletion, export, and consent controls where applicable.
- Protect unpublished partner content and culturally restricted records with row-level authorization.
- Maintain content revision, moderation, and AI-response audit trails.
- Create incident response, backup, restore, and disaster-recovery procedures before public launch.

14. Arizona Explorer MVP Deployment

14.1 Initial Content Scope

- Arizona as a root region within Earth.
- Five to ten flagship places, including the Grand Canyon and selected cities, museums, parks, or observatories.
- A curated set of species, events, organizations, artifacts, and concepts connected to those places.
- At least three timeline lenses.
- One complete learning quest and one career pathway.
- An AI Guide restricted to the published Arizona knowledge base plus approved sources.

14.2 Suggested Infrastructure

Capability	MVP choice
Web hosting	Managed Next.js hosting or container platform

API	FastAPI container
Database	Managed PostgreSQL with PostGIS and vector extension
Media	Cloud object storage with CDN
Authentication	Managed identity provider
Maps	Mapbox or Cesium depending globe requirements
AI	OpenAI API with retrieval orchestration
Observability	Central logs, metrics, error tracking, and AI quality traces
CI/CD	GitHub-based automated test and deployment pipeline

14.3 MVP Quality Gates

- All primary journeys work using keyboard only.
- Every published factual page has sources and a fact-class label.
- AI answers cite retrieved evidence or clearly state that evidence is insufficient.
- Map content has a nonvisual alternative.
- Core pages meet agreed performance budgets on mobile and desktop.
- Rights and attribution records exist for every media asset.
- Backups and restoration are tested.
- Editorial corrections can be published without code changes.

15. Phased Evolution

Gate	Architecture expansion
Gate 1: Foundation	Vision, PRD, architecture, knowledge model, design system, governance.
Gate 2: Arizona Explorer	Map, places, timelines, AI Guide, quest, Forge Passport.
Gate 3: Learning Forge	Educator tools, portfolios, deeper skills and career pathways.
Gate 4: Creator Forge	Original and licensed story tools, Graphic Novel Studio, world-building tools.
Gate 5: Living Worlds	Richer simulation, 3D places, museums, zoos, farms, schools, hospitals, starships.
Gate 6: Immersion	AR, VR, mixed reality, multiplayer, adaptive AI companions.
Gate 7: WorldBuilder Network	Partner publishing, creator marketplace, institutional APIs, global contributor governance.

Appendix A — Example Arizona Knowledge Graph

Subject	Relationship	Object
Arizona	contains	Grand Canyon National Park
Grand Canyon National Park	contains	Grand Canyon
Colorado River	flows_through	Grand Canyon
Grand Canyon	formed_by	Colorado River and geological processes
California condor	observed_at	Grand Canyon region
Grand Canyon Railway	serves	South Rim area
Grand Canyon	associated_with	multiple Tribal Nations and communities
Grand Canyon geology quest	teaches	erosion, stratigraphy, deep time, and stewardship
Grand Canyon geology quest	practices	observation, evidence evaluation, and map reading
Geologist	requires	earth science, field observation, data analysis, and communication skills

Note: community-associated content should be developed with appropriate partners and should not imply that distinct Tribal Nations share a single history, identity, or perspective.

Appendix B — MVP API Surface

Endpoint	Purpose
GET /worlds	List available worlds.
GET /places	Search/filter places by map bounds, type, or parent.
GET /places/{id}	Return place page data, sources, accessibility presentation, and related entities.
GET /entities/{id}/relationships	Traverse typed relationships.
GET /timelines/{id}	Return ordered events and uncertainty metadata.
POST /guide/questions	Submit a grounded AI Guide question with context and presentation preferences.
GET /quests/{id}	Return learning objectives, steps, resources, and accessibility variants.
POST /passport/entries	Save a visit, completed activity, reflection, badge, or skill evidence.
GET /search	Unified keyword, semantic, geographic, and faceted discovery.
POST /admin/entities	Create editorial draft content.
POST /admin/review/{id}	Approve, reject, annotate, or return content for revision.

Appendix C — Glossary

Term	Definition
Entity	A distinct thing represented in the knowledge

	model.
Relationship	A typed, sourced connection between two entities.
Claim	A factual or interpretive assertion supported by evidence or explicitly labeled otherwise.
Fact class	A label distinguishing documented fact, interpretation, tradition, mythology, fiction, adaptation, simulation, and user-created material.
Canon scope	The continuity or version in which fictional information applies.
Provenance	The origin, ownership, processing history, and review record of content.
Presentation	An audience- and accessibility-specific rendering of shared knowledge.
Journey	A curated or adaptive sequence of places, entities, activities, and questions.
Forge Passport	The learner-controlled record of exploration, projects, achievements, skills, and evidence.
Gate	A major product stage or release family.
Key	A principle or capability that unlocks a Gate.